

Capstone controlled experiments

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Trace matched two-update learning-rate experiments

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

Look for: Write a testable capstone hypothesis and a credible baseline

Reference example, not the only acceptable wording: Teacher explanation: Predict $\max(0, a \cdot x)$ without bias; baseline $a = 1$; candidate starts at 1, learning rate 0.1, two full-batch updates on T1/T2. Hypothesis: candidate validation MSE on V1/V2 is strictly below the baseline MSE.

Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate throughput and amortised time per example from repeated timings

Reference example, not the only acceptable wording: Teacher explanation: A: 2.0 seconds, 20 ms per example, 50 examples per second. B: 1.2 seconds, 12 ms per example, about 83.33 examples per second. The boundary covers 100 completed sequential predictions on preprocessed inputs, excluding loading and preprocessing. It does not establish worst-case or percentile request latency.

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a reused feature extractor and a newly fitted output head

Reference example, not the only acceptable wording: Teacher explanation: The extractor gives $h = 6$ and the head predicts 1. Weights $[1, 1]$ and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or calibration.

Use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in training and evidence, not a real-world deployment. Preserve written plans, data roles and raw predictions. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

Outcomes

Trace matched two-update learning-rate experiments

Select a candidate using a prespecified validation rule

Explain the limits of a one-factor comparison

Preparation

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

Prerequisite answer

Data and splits, initial parameters, architecture, loss, update count and evaluation procedure must stay fixed; change only the intended learning rate.

End goal for the pictorial case

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Continue with training (1,2) and (2,4), validation (3,6) and (4,8), and prediction $\max(0, a \cdot x)$. Compare learning rates 0.1 and 0.2. Each run starts at $a = 1$, has no bias and takes two full-batch updates. Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not. Select lower validation MSE, with a tie going to learning rate 0.1.

For positive weights and these positive inputs, the ReLU is active. Training MSE is $((a-2)^2 + (2 \cdot a - 4)^2)/2 = 2.5 \cdot (a-2)^2$. Its gradient is $5 \cdot (a-2)$. Update a to $a - \text{rate} \cdot 5 \cdot (a-2)$. At rate 0.1, weights are 1.5 then 1.75. At rate 0.2, weights are 2 then 2. The formula assumes the active region; do not extend it blindly to nonpositive weights.

Term · Meaning

Term: Controlled comparison

Meaning: A comparison holding specified conditions constant while changing an intended factor

Term: Full batch

Meaning: An update whose gradient uses all training rows

Term: Selection rule

Meaning: A rule for choosing a candidate using development evidence

Term: Confounder

Meaning: A differing condition that prevents isolating the intended explanation

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Same initial $a = 1$

Procedure: Rate 0.1, two updates

Permitted conclusion: Validation MSE 0.78125

Plan or input: Same initial $a = 1$

Procedure: Rate 0.2, two updates

Permitted conclusion: Validation MSE 0

Plan or input: Same validation rows

Procedure: Choose lower MSE

Permitted conclusion: Select rate 0.2

The selected run fits this exact constructed relationship. Its zero error does not establish universal reliability.

Guided: $2.5 - 0.3 \cdot 2.5 = 1.75$; validation MSE 0.78125. Equal final weights here give equal final predictions, but the intermediate weights differ. Exit: the score compares complete procedures with different budgets, not an isolated learning-rate effect; match update counts and all other settings.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

At $a = 1.75$, validation predictions are 5.25 and 7. Errors -0.75 and -1 square to 0.5625 and 1, giving MSE 0.78125. At $a = 2$ the predictions equal 6 and 8, so MSE is zero. Rate 0.2 wins the written validation rule. This selection uses two validation cases, not four independent cases because two runs used them.

If one run used extra epochs or different training rows, the comparison would mix factors. Either rerun a matched experiment or report the differences and avoid claiming an isolated learning-rate effect. A zero

validation score is also compatible with a very simple task and says nothing about a new distribution.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone controlled experiments

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

The facts we will use

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Controlled toy-pattern comparison

correct labels out of 6 shared patterns



Bar length shows the supplied value; the marked baseline is zero.

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify the intervention

What happens and why: B adds the centre-feature connection; all other declared settings stay fixed.

Step: 2. Count outcomes

What happens and why: A is 4/6 correct; B is 5/6.

Step: 3. Inspect paired cases

What happens and why: Only case three changes correctness; case five fails for both.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

What this example does not establish

These descriptive toy results do not establish significance, stable superiority or the full learned mechanism.

Change one thing in the pictured case

Change: Change B's training set as well as the feature connection.

Prediction: The experiment no longer isolates the connection change.

Reason: A training-data difference supplies an alternative explanation for the observed result.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and ask students to explain the data roles.

Time or stage: 7 to 18 minutes

Teacher action: Read section 1 and complete W1 with a recorded reason.

Time or stage: 18 to 32 minutes

Teacher action: Trace section 2 and complete W2; compare raw values before conclusions.

Time or stage: 32 to 45 minutes

Teacher action: Explain the diagram and complete W3 in pairs; record any support.

Time or stage: Session 2 start to 8 minutes

Teacher action: Complete the guided response and discuss why the evidence has limits.

Time or stage: 8 to 20 minutes

Teacher action: Read the remaining explanation and prepare a concise experiment record.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently. Require calculations and a bounded claim.

Time or stage: 35 to 45 minutes

Teacher action: Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw predictions, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns. Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Reduce help deliberately

Completion task: Compute both updates for rates 0.1 and 0.2 using gradient $5^*(a-2)$.

Teacher check: Teacher explanation: Rate 0.1: 1 $\rightarrow$ 1.5 $\rightarrow$ 1.75. Rate 0.2: 1 $\rightarrow$ 2 $\rightarrow$ 2. Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Reusing two rows across runs leaves two distinct validation cases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?

Reasoned teacher answer

A has four correct, B five; case three explains the count gain and case five fails for both. Changing training data alongside the connection entangles two interventions, so the gain cannot be attributed to the connection alone.

Changed-case reasoning: The experiment no longer isolates the connection change. A training-data difference supplies an alternative explanation for the observed result.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

More runs mean more validation examples: Reusing two rows across runs leaves two distinct validation cases.

The largest rate must be best: The active-region update may overshoot; results depend on task, rate and budget.

Zero validation error proves the final result: Final rows remain unopened and the selected candidate has not yet been evaluated on them.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner

restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: $2.5 - 0.3 \cdot 2.5 = 1.75$; validation MSE 0.78125. Equal final weights here give equal final predictions, but the intermediate weights differ. Exit: the score compares complete procedures with different budgets, not an isolated learning-rate effect; match update counts and all other settings.

W1 Name the controls

List six settings held fixed and the single intended changed factor.

Teacher explanation: Hold training and validation rows, initial a , architecture/bias policy, loss, update count and evaluation metric fixed. Change learning rate.

W2 Trace both runs

Compute both updates for rates 0.1 and 0.2 using gradient $5 \cdot (a-2)$.

Teacher explanation: Rate 0.1: 1 → 1.5 → 1.75. Rate 0.2: 1 → 2 → 2. Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

W3 Select without extra evidence

Recompute validation MSE for the final weights and apply the written selection rule. How many distinct validation rows were used?

Teacher explanation: MSE 0.78125 for $a = 1.75$ and 0 for $a = 2$, so select rate 0.2. There are two distinct validation rows; re-evaluation does not add cases.

W4 Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Teacher explanation: Rate 0.1: gradients -10,-5; weights 2,2.5. Validation errors -1.5,-2 give MSE 3.125. Rate 0.2: weights 3,3, errors 0,0 and MSE 0. Select 0.2, with only validation evidence on two constructed rows.

W5 Audit a mixed comparison

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different training rows. Can the difference be attributed to rate alone? Propose a repair.

Teacher explanation: No: update count and training data also differ. Rerun both rates with the same initial parameters, architecture, rows, loss, two-update budget and evaluation procedure; preserve and label the earlier mixed comparison.

Independent assessment and fresh retry

W4 requires weights 2.5 and 3, MSE 3.125 and 0, selection 0.2 and a validation-only conclusion. Fresh retry: targets are $4*x$ on the same inputs, gradient $5*(a-4)$. Final weights are 3.25 and 4; validation MSE 7.03125 and 0. Require the same data-role limit.

Assess independent transfer

Selected case: For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Selected case answer

Teacher explanation: Rate 0.1: gradients -10,-5; weights 2,2.5. Validation errors -1.5,-2 give MSE 3.125. Rate 0.2: weights 3,3, errors 0,0 and MSE 0. Select 0.2, with only validation evidence on two constructed rows.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5 \cdot (a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw predictions, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns. Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Week 35 evaluates a frozen candidate and records limitations before considering revision.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

7. Prepare, teach and assess

Teaching target: Trace matched two-update learning-rate experiments

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Compute both updates for rates 0.1 and 0.2 using gradient $5 \cdot (a-2)$.

Reasoned practice answer: Teacher explanation: Rate 0.1: $1 \rightarrow 1.5 \rightarrow 1.75$. Rate 0.2: $1 \rightarrow 2 \rightarrow 2$. Gradients are -5 then -2.5 for the first run and -5 then 0 for the second.

Check every learner before discussion: Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Reusing two rows across runs leaves two distinct validation cases.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change B's training set as well as the feature connection. Require a prediction and a reason before comparing. Expected result: The experiment no longer isolates the connection change. Reason: A training-data difference supplies an alternative explanation for the observed result.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.